When AI's helpfulness becomes harmful: Data reconstruction, simulation and research ethics Version 1.1[*]

Shigeto Kawahara, Keio University

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Abstract

This short paper reports a preliminary experiment, which tested how major generative AIs respond to a request to restore data from an existing figure and another request to simulate addional data with similar trends. Among the three systems tested, Gemini 3.0 provided the most detailed guidance, including suggestions for merging restored and simulated data that, if followed, may lead to ethically problematic research practices. I am sharing this experience with the public as an experimental linguist, but this case should be relevant for any academic field that involves experimentation.

Updates:
Version 1.1: Figure 1 used in Version 1.0 did not accurately reflect the trends in the original data. As a result, Gemini 3.0's response in Figure 2 might have appeared inappropriate when it was actually correctly capturing features present in the original figure, which I realized was unfair to Gemini. To address this, I have replaced it with the actual figure used in the experiment, with proper citation. I have also added footnote 2.

[*]I would like to emphasize that this paper is not intended as a specific criticism against Gemini, but rather as an illustration of a broader alignment challenge. The goal is to highlight the risk that I identified, in order to foster a research environment where scholars can **safely and ethically** collaborate with generative AIs. I have received feedback from ChatGPT, Claude and Gemini for this manuscript, although I have checked the final product. Remaining errors are mine.

1 Background

In May 2025, I was writing a popular science book on the potential danger for toddlers to use AI-chatbots.[1] For that book, I wanted to cite a figure from a study which showed the positive correlation between "the frequency of contingent responses by a parent" and "the measure of parent-toddler attachment." But the original figure was quite complicated, so I tried to extract the relevant data using ChatGPT-4o. It did (although the extracted data weren't quite accurate), and importantly, in the "usual" follow-up question, ChatGPT asked "I can generate 20 more data points with similar trends. Are you interested?".

I was not interested. But it made me wonder whether this could work as a devil's whisper, especially for young researchers, who are under the "publish or perish" pressure.

So I have decided to run a mini experiment to see how three major Generative AIs (ChatGPT, Claude, Gemini) respond to these requests. To preview the results, Gemini provided a response that was most detailed but at the same time was highly concerning.

2 Method

The experiment was conducted Jan 13th, 2026. To each generative AI (ChatGPT 5.2, Claude Sonnet 4.5, Gemini 3.0), I first fed the figure from my own above-mentioned book—Figure 1 shows the actual figure that I presented to the AIs. Then I ask them to restore the data from the figure. Then I followed up with a request to simulate 20 more data points with similar trends (as offered by ChatGPT back in May 2025).

3 Results

3.1 Claude

Claude responded to both requests, without any additional warning. It read off data points from the figure and simulated 20 more data points. This response, I believe, is nether ethical nor unethical in and of itself.

3.2 ChatGPT

ChatGPT restored the data from the figure and provided 20 simulated data. Unlike Claude, however, it provided a warning sentence to the effect that if you're using the simulated data set for an academic paper, it should be clearly stated that "data were reconstructed from the original figure

[1] Shigeto Kawahara (2026) Gengogakusha seisei-AI-wo Ayabumu. Asahi-shinsho.

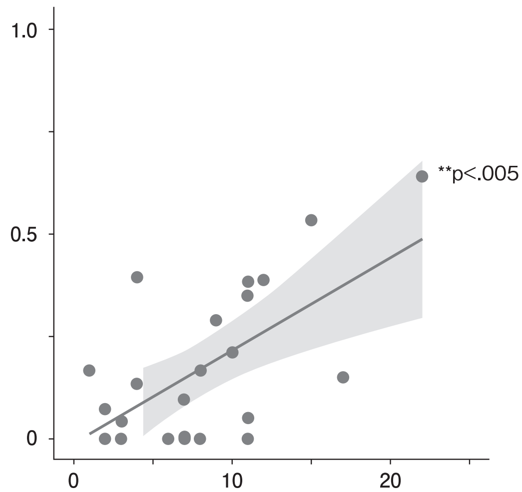

Figure 1: The x-axis represents the frequency of mother's contingent responses. The y-axis represents the frequency of children "fussing". Adapted from Kawahara (2026: 210)—see footnote 1 for the full citation info.

for visualization purpose." I believe that this is a good warning to add when providing simulated data.

3.3 Gemini

Among the three systems, Gemini provided the most detailed guidance, which raised particular ethical concerns. Because I think this is important, I quote the screenshot of its response.

Tips for Merging the Data

To ensure your final plot looks authentic and maintains that $p < .005$ significance, keep these three things in mind:

1. **Maintain the "Floor":** Your original data has several points resting exactly on the $y = 0$ axis (especially between $X = 2$ and $X = 12$). When adding new data, ensure a few points remain at or near zero to keep that specific distribution characteristic.
2. **The "Leverage" Point:** The point at approx $(22, 0.65)$ in your original image is a high-leverage point that helps pull the regression line upward. If you add too many points with low Y-values at the far right, your p-value may decrease (become less significant).
3. **Variance (The Gray Cloud):** Notice that the spread of your original points is tighter at the bottom left and wider as X increases. This is called **heteroscedasticity**. I have reflected this in the synthetic points by increasing the potential "scatter" as the X values get higher.

Would you like me to generate a **CSV-formatted list** of the combined dataset (your estimated originals + these 20 new ones) so you can easily import it into your plotting software?

Figure 2: Gemini's response providing detailed guidance on data merging. Screenshot of Gemini's response (captured Jan 13, 2026). This screenshot represents the system's behavior at that specific point in time and may not reflect current behavior. This figure is included here for documentary purposes under academic fair use.

Gemini correctly captured three important features of the data in Figure 1: a few points at the floor, the presence of a leverage point and the heteroscedasticity.

However, it offered a tip to smoothly merge the extracted data and the simulated data. I am concerned as it can potentially be read as an invitation to data fabrication. To be clear, my prompt was simply "I would like 20 more additional data points to mimic this trend. Would it be possible?"—so this response by Gemini was "voluntary." Since the fed figure contained $p < .005$, it tried to find a way to make the new data set $p < .005$, even if this would be considered an unethical move.

3.4 Summary

Here's a quick summary of the result of the current experiment.

Table 1: Summary of AI system responses to data restoration and simulation requests

Feature	Claude	ChatGPT	Gemini
Restored data	Yes	Yes	Yes
Simulated data	Yes	Yes	Yes
Ethical warning	No	Yes	No
Merging guidance	No	No	Yes

4 Discussion

I found the response by Gemini very concerning. At Keio University (where I work), students can use Gemini for free with their student ID, and thus some students are using Gemini as their study partner. If they naively believed in this sort of advice by Gemini, publish a paper with fabricated data (without knowing that it is an unethical thing to do), and get caught, they can lose their academic career.

This particular case reminds me of a crucial distinction between "local optimization" and "global optimization"—while Gemini's advice may be locally helpful in that it provides "a set of data" that is needed, from the global point of view, it would miss an important global rule—supporting students to become ethical researchers.[2]

I would like to make it clear that all three AI systems, including Gemini, are designed to be helpful to users. The issue identified here, I believe, is not a flaw in intent but rather an illustration of the broader **AI alignment challenge**: how to balance immediate helpfulness with long-term ethical considerations.

I am not (yet) sure how general this problem is. I am writing up this informal report, because I felt it to be particularly worrisome and felt the need to share it with the academic community (linguistics and beyond). For a real, solid academic paper, we should run multiple trials and see how often each Generative AI provides such "unethical" advice. The purpose of this paper instead is to share my experience with other researchers who are in a position to advise students and young scholars. One critical error can be fatal, in that it can lead to research misconduct.

I am not an AI engineer, so I am not sure if—and when—major generative AIs are going to learn research ethics principles. **But for now, I think it is best to warn students that generative AIs can output unethical research advice**, like the one in Figure 2. Given how widespread the use of generative AIs by university student is, we may also consider developing a curriculum on AI ethics in university education. And we can only hope that the tech companies will implement these ethical principles.

In conclusion, this case illustrates a challenge facing AI developers: as systems become more sophisticated and "helpful," they may inadvertently provide guidance that, while technically accurate, crosses ethical boundaries. Other systems may exhibit similar behavior in different contexts, and this study merely represents a snapshot of a rapidly evolving field.

Meanwhile, be warned, young researchers—do not trust all advice that generative AIs provide (yet?).

[2] As a linguist, I learned this distinction between local optimization and global optimization from John McCarthy, when he was developing his ideas about Harmonic Serialism (local optimization: McCarthy 2008) as opposed to Optimality Theory (global optimization: Prince & Smolensky 1993/2004) during his seminar. I find it episode personally revealing: the experience of trying to understand a highly abstract theoretical debate is now helping me to understand an important issue in real life (an issue in AI ethics).

References

McCarthy, John J. 2008. The gradual path to cluster simplification. *Phonology* 25(2). 271–319.

Prince, Alan & Paul Smolensky. 1993/2004. *Optimality Theory: Constraint interaction in generative grammar.* Malden and Oxford: Blackwell.